

10-point endodontic outcomes checklist

Use this chairside checklist when outcome risk needs to be made visible. The aim is controlled diagnosis, infection control, apical management, shaping, obturation, restoration, review, and retreatment judgment.

01

DIAGNOSE AND PLAN

Name the baseline risk

Outcome confidence starts with the pre-operative diagnosis, not the file sequence.

Healing rates are influenced by the starting condition of the tooth. A tooth with a pre-operative lesion, symptoms, retreatment history, cracks, or poor restorability carries a different risk profile from a vital primary case. Record pulpal and periapical diagnosis, lesion size, restorative status, periodontal context, and patient factors before the technical plan is chosen. This keeps the outcome conversation honest and gives the clinician a baseline for later review.

[YouTube: Endodontics | Pulpal and Periapical Diagnoses](#)

02

DIAGNOSE AND PLAN

Control infection before speed

Endodontic success is biological healing supported by technical control.

A fast preparation is not the same as an effective preparation. Instrumentation creates access for irrigation, removes infected dentine where possible, and supports obturation, but the clinical aim is infection control. Maintain patency where appropriate, clean flutes, refresh irrigant, and avoid debris compaction. When the canal becomes blocked, ledged, or over-enlarged, the biological objective becomes harder to achieve.

[YouTube: How to perform successful endodontic treatment](#)

03

CONTROL THE APICAL THIRD

Respect the apical limit

Apical extent is one of the clearest technical variables linked to healing.

The existing Evidence page cites outcome literature supporting controlled root filling close to the radiographic apex, with the apical constriction respected. Ricucci and Siqueira reported higher healing when treatment remained approximately 0.5-1.5 mm short of the apical foramen, while overfilling and excessive shortness were associated with poorer histologic outcomes. Use working length evidence, tactile feedback, and apical-control discipline together rather than treating length as a single number.

[YouTube: Apical control in Endodontics - Dr Heinrich Dippenaar \(WEB70\)](#)

04

CONTROL THE APICAL THIRD

Preserve the original pathway

Procedural error converts a treatable case into an outcome risk.

Transportation, ledging, perforation, zipping, or file separation can reduce healing predictability by changing the prepared pathway or blocking disinfection. Small-file scouting, reproducible glide path confirmation, short engagement, and careful feedback protect the original canal course. Avatar Tip and Transform Technology fit here: they are not outcome claims by themselves, but they support the clinical behaviour that outcome evidence rewards: centred advancement, control in curvature, and respect for the apical third.

[YouTube: How to Decide Master Apical Size in Root Canal Treatment](#)

05

SHAPE, CLEAN, AND SEAL

Shape for irrigation and seal

Preparation quality matters because it creates the conditions for cleaning and obturation.

Outcome evidence links obturation quality to prognosis, but obturation quality is built earlier. A controlled preparation creates space for irrigant exchange, removes restrictive interference, and gives the filling material a stable pathway. The clinical balance is important: too little shape limits cleaning and seal, while too much shape removes dentine and increases procedural risk. The target is a reproducible preparation that serves disinfection and obturation rather than a shape created for appearance alone.

[YouTube: Elements of perfect obturation](#)

06

SHAPE, CLEAN, AND SEAL

Obturate densely without extrusion

Dense obturation supports healing when it is paired with apical control.

A radiographically dense fill is useful only when it is biologically positioned and supported by cleaning. Voids, short fills, and overextension each signal different failure risks. The apical stop, canal taper, irrigation sequence, and final cone fit all affect whether obturation will seal predictably without extrusion. Use the fill as a final check on the preparation path rather than as an isolated technical endpoint.

[YouTube: Obturation & Restoration](#)

07

SHAPE, CLEAN, AND SEAL

Restore the tooth promptly

Long-term survival depends on endodontics and restoration working together.

The existing page cites four-year tooth survival of approximately 95% in prospective outcome work, but survival is not achieved by canal treatment alone. Coronal leakage, cuspal fracture risk, inadequate ferrule, and delayed restoration can undermine an otherwise controlled endodontic procedure. Plan the restorative pathway before treatment, seal promptly, and review whether the tooth is protected from reinfection and structural failure.

[YouTube: Video Lecture : Restoration Of Endodontically Treated Tooth](#)

08

REVIEW AND DECIDE

Review healing over time

A persistent radiolucency needs context before retreatment is chosen.

Apical healing can take time, and a radiograph is a moment in a longer biological process. Compare symptoms, percussion, sinus tract status, lesion size, periodontal findings, restoration quality, and previous baseline images. Do not retreat only because a radiolucency remains visible if the clinical picture is improving. Equally, do not ignore persistent symptoms, enlarging lesions, coronal leakage, or missed anatomy. Review is a structured decision, not a reflex.

[YouTube: Not every tooth with a periapical radiolucency needs endodontic retreatment](#)

09

REVIEW AND DECIDE

Treat failure as cause-finding

Retreatment is most controlled when the reason for persistence is defined.

Persistent apical periodontitis should trigger a cause-finding sequence. Check for missed canals, inadequate access, short or overextended filling, coronal leakage, separated instruments, ledges, perforation, cracks, periodontal involvement, and restorative failure. CBCT may be useful when the cause is not visible on periapical radiographs. Retreatment should be entered with a clear hypothesis and an exit plan, including referral when visibility, anatomy, or restorability makes control uncertain.

[YouTube: Endodontic Retreatment Explained](#)

10

REVIEW AND DECIDE

Link survival to the whole pathway

The outcome is a retained, restorable, symptom-free tooth in function.

Healing percentages and survival percentages describe different endpoints. Primary and secondary treatment can both produce high healing rates when case selection, infection control, apical management, obturation, and restoration are controlled. Tooth survival also depends on structure, periodontal status, occlusion, restoration, and patient maintenance. Use the full pathway as the outcome frame: diagnose, disinfect, shape, seal, restore, review, and decide early when referral protects the tooth.

[YouTube: Endodontic Treatment Outcomes](#)

SOURCE BASE

The Evidence checklist combines the existing EndoTech outcomes page with outcome literature and a 20-video YouTube review focused on endodontic diagnosis, apical periodontitis, apical control, obturation, restoration, retreatment, and long-term tooth survival. The checklist uses those sources to support clinical decision-making around controllable outcome variables, not to claim that any file system alone determines healing.

- EndoTech NZ existing Clinical Evidence page, including the Ng UCL Eastman outcome summary and the Ricucci/Siqueira apical-limit reference.
- Ng YL. UCL Eastman outcome research in primary and secondary non-surgical root canal treatment; meta-analysis and prospective cohort summary as cited in the existing EndoTech Evidence page.
- Ricucci D, Siqueira JF Jr. Apical limit of root canal instrumentation and obturation in teeth with necrotic pulps and apical periodontitis: histologic analysis. *Oral Surgery, Oral Medicine, Oral Pathology, Oral Radiology, and Endodontology*. 2011;112(6):825-842.
- EndoTech NZ curated YouTube outcomes review: 20-video set gathered with the youtube-researcher workflow on 2026-06-02. Local research file: tmp/youtube-endodontic-outcomes-curated.json.
- American Association of Endodontists. Endodontic Retreatment Explained. YouTube.
- NIOM. How to perform successful endodontic treatment. YouTube.
- South African Dental Association. Apical control in Endodontics - Dr Heinrich Dippenaar (WEB70). YouTube.